

AegisCare HMS / LabTrack v5.5

System	AegisCare HMS / LabTrack v5.5
Supabase project ref	tgynjasgnerucrlwedui
Base URL (Supabase)	https://tgynjasgnerucrlwedui.supabase.co
Base URL (DHA HIE)	https://ilm-dev.dha.go.ke/uat-middleware/api/v1
Document status	DRAFT for certification submission
Date	2026-08-12

Section 1 — API Overview

AegisCare exposes four API surfaces:

Surface	Base	Auth	Consumers
Supabase REST (PostgREST)	https://tgynjasgnerucrlwedui.supabase.co/rest/v1/{table}	Supabase JWT (anon key or user REST/RLS)	Frontend (via @supabase/supabase-js), server functions
Supabase Edge Functions	https://tgynjasgnerucrlwedui.supabase.co/functions/v1/{fn}	Supabase JWT (verify_jwt = true, anon key or user)	Frontend, DHA/SHA integrations, certification tooling
Supabase RPCs	.../rest/v1/rpc/{fn}	Supabase JWT + RLS/GRANT	Frontend (e.g., generate_fhir_encounter, not with 305 request)
DHA HIE (external target)	https://ilm-dev.dha.go.ke/uat-middleware/api/v1	OAuth2 POST /tenants/token	PENDING credentials

All traffic is HTTPS. Response payloads for FHIR endpoints are served as application/fhir+json.

Section 2 — Authentication

2.1 Supabase Auth (in force)

- Anonymous key (publishable): sb_publishable_... — set as VITE_SUPABASE_PUBLISHABLE_KEY; used client-side; RLS restricts it to authenticated-approved data only.
- User session JWT: obtained from Supabase Auth (email/password); sent as Authorization: Bearer <token>; powers RLS (auth.uid(), is_approved()).
- Service role key: server-side only (src/integrations/supabase/client.server.ts, supabaseAdmin) — never exposed to the browser; used by admin server functions behind requireSupabaseAuth + assertAdmin.
- Edge function calls: Authorization: Bearer <user JWT or anon key> + apikey header; platform enforces verify_jwt = true.

2.2 DHA OAuth2 (external — pending)

- Endpoint: POST https://ilm-dev.dha.go.ke/uat-middleware/api/v1/tenants/token
- Body (JSON): { "client_id": "...", "client_secret": "..." }

- Response: { "access_token": "...", "expires_in": ..., "token_type": "..." }
- Status: ☐ PENDING — credentials require DHA developer-account onboarding (<https://developer.dha.go.ke>); facility headers X-Facility-Id (FR code) and X-Facility-Id-Type: fr-code apply to service-account calls (per hie-docs.dha.go.ke).

Section 3 — Edge Functions API Reference

Common conventions:

- URL: <https://tgynjasgnerucrlwedui.supabase.co/functions/v1/{name}>
- Method: POST (JSON body); OPTIONS supported (CORS preflight).
- Headers: Authorization: Bearer <jwt>, apikey: <key>, Content-Type: application/json.
- All functions: Authentication required (platform JWT verification).

3.1 `claims-dispatcher`

Endpoint	/functions/v1/claims-dispatcher
Method	POST
Authentication	Required (JWT + supabase.auth.getUser() in-function)
Status	Live (stub mode) — queues locally; external API calls pending credentials

Request body:

```
{
  "encounter_id": "uuid",
  "patient_id": "uuid",
  "insurer_type": "sha_shif | private | corporate | cash | null",
  "trigger": "encounter_closed | claim_submit | fhir_sync | manual"
}
```

Response (200):

```
{
  "success": true,
  "encounter_id": "uuid",
  "results": [
    {
      "queue_type": "fhir_sync | sha_claim | private_claim | cash_receipt",
      "handler": "FhirSyncHandler | ShaClaimsHandler | PrivateClaimsHandler | CashReceiptHandler",
      "status": "queued | skipped",
      "queue_id": "uuid | null",
      "message": "..."
    }
  ],
  "note": "STUB MODE — All submissions queued locally. External API calls activated in Phase 3."
}
```

Errors: 400 (missing encounter_id/patient_id), 401 (missing/invalid auth), 500 (handler error). Behaviour: FhirSyncHandler checks patient_consent.hie_data_sharing_consented and marks fhir_sync as skipped when consent is absent; SHA/private/cash routes by insurer_type.

3.2 `fhir-patient`

Endpoint	/functions/v1/fhir-patient
Method	POST
Authentication	Required
Status	Live

Request: { "patient_id": "uuid" } → Response 200 application/fhir+json:
 FHIR R4 Patient (id, meta.profile, identifier [facility file number],
 name, gender, birthDate, telecom, address, deceasedBoolean,
 managingOrganization). 400 missing id; 404 unknown patient; 500 server error.

3.3 `fhir-encounter`

Endpoint	/functions/v1/fhir-encounter
Method	POST
Authentication	Required
Status	Live

Request: { "encounter_id": "uuid" } → Response 200 application/fhir+json:
 FHIR R4 Encounter (status, class AMB/IMP/EMER, priority for
 emergencies, subject, period, diagnosis[] with Condition/{id}
 references and rank, serviceProvider KMHFL identifier,
 hospitalization for referrals out). 400/404/500 as above.

3.4 `fhir-condition`

Endpoint	/functions/v1/fhir-condition
Method	POST
Authentication	Required
Status	Live

Request: { "encounter_id": "uuid" } → Response 200 application/fhir+json:
 array of FHIR R4 Condition resources from encounter_diagnoses
 (verificationStatus mapped from diagnosis_type, ICD-11 code.coding,
 subject, encounter, recordedDate, note, sequence extension). Empty
 array when no diagnoses. 400/404/500 as above.

3.5 `icd11-search`

Endpoint	/functions/v1/icd11-search
Method	POST
Authentication	Required
Status	Live — returns 500 with "ICD API credentials not configured" until ICD_CLIENT_ID/ICD_CLIENT_SECRET secrets are set; frontend falls back to local icd11_codes

Request: { "query": "string (≥2 chars)" } → Response 200:

Section 4 – DHA HIE API Integration

(all endpoints per the DHA HIE API catalogue, <https://hie-docs.dha.go.ke>).

Integration seam: all DHA-bound work is queued through `dha_outbound_queue` (queue_type CHECK incl. `fhir_sync`, `sha_claim`, `shr_access_notification`) with `shr_transmission_log` recording every transmission attempt — the queue becomes the live submitter when credentials arrive, with no application-UI changes required.

Section 5 — Database Functions Reference

Function	Signature	Purpose	Grants
accrue_daily_bed_charges()	() → void	Inserts invoice_line_items (bed_day) for admitted admissions; cron accrue-daily-bed-charges 21:01 UTC	cron

Function	Signature	Purpose	Grants
accrue_daily_mortuary_charges()	() → void	Accrues daily mortuary storage charges (invoice line or total_storage_charges); cron accrue-daily-mortuary-charges 21:31 UTC	cron
generate_fhir_encounter(p_encounter_id uuid)	(uuid) → jsonb	FHIR R4 Encounter builder (Second Schedule coverage); SECURITY DEFINER, search_path=public	authenticated (not indicated from anon)
enforce_encounter_lock()	() → trigger	Blocks INSERT/UPDATE/DELETE on child records of signed/finalized/completed encounters (attached to encounter_diagnoses, clinical_notes, radiology_orders)	trigger
auto_generate_sha_claim()	() → trigger	On encounters.status → 'signed'; (insurance): creates draft sha_claims + seeds sha_claim_items from invoice line items	trigger (SECURITY DEFINER)
enforce_insurance_visit_limit()	() → trigger	Blocks insurance_covered above per_visit_limit for fixed_per_visit / percentage_with_cap insurers	trigger
archive_old_audit_logs()	() → void	Moves audit_log rows older than 2 years to audit_log_archive; logs to audit_archive_runs; cron archive-audit-logs-nightly 23:00 UTC	cron (SECURITY DEFINER)
is_approved(_user_id uuid)	(uuid) → boolean	RLS gate — all 19 roles	authenticated
has_role(_user_id uuid, _role app_role)	→ boolean	Role check	authenticated
user_has_permission(_user uuid, _perm text)	→ boolean	Permission check via role_permissions	authenticated
get_moh_705_report(p_start date, p_end date, p_form_type text)	→ table	MOH 705 A/B morbidity counts	authenticated
validate_and_get_icd11(search_code text)	(text) → (p_code, p_title, p_uri, p_is_cached)	ICD-11 validation/resolution against icd11_codes	internal/trigger
get_contracted_price(item_type, item_id, insurer_id)	→ numeric	Contracted rate-card lookup (contracted_prices)	authenticated
log_break_glass_access(p_patient_id uuid, p_justification text, p_accessed_by uuid, p_accessor_email text)	() → void	Break-glass audit + SHR access notification	authenticated
verify_patient_identity(...) / verify_practitioner(...)	stubs	IPRS / HWR verification stubs (return stub data until credentials)	authenticated
send_lab_result_to_room(p_registration_id uuid, p_room_id uuid) / send_lab_results_to_requesting_room(p_encounter_id uuid)	→ uuid	Route patient back after lab results	authenticated
send_radiology_result_to_room(...) / send_radiology_results_to_requesting_room(...)	→ uuid	Radiology routing equivalents	authenticated
refresh_moh_aggregates(target_month date)	() → void	Recomputes moh_monthly_aggregates	admin/reporting
create_invoice_for_encounter() / recalc_invoice_totals() / recalc_invoice_payments() / sync_invoice_*	triggers	Invoicing integrity chain	triggers
route_registration_billing() / route_registration_to_lab() / route_registration_to_service_room() / route_prescription_to_pharmacy() / get_return_room()	routing	Queue/room automation	triggers/RPC

Function	Signature	Purpose	Grants
set_sha_fund_type()	trigger	Auto PHF/SHIF/ECCIF classification on encounter insert	trigger
dispense_prescription_stock()	trigger	Stock deduction + negative-stock guard on dispense	trigger
trg_medication_dispense_fhir()	trigger	FHIR MedicationDispense → dha_outbound_queue	trigger (SECURITY DEFINER)
trg_shr_access_notification()	trigger	SHR access alert queueing	trigger (SECURITY DEFINER)
trg_shr_transmission_log() / _insert()	triggers	SHR transmission metadata log	triggers (SECURITY DEFINER)
sync_encounter_diagnoses_from_jsonb()	trigger	encounters.diagnoses JSONB → encounter_diagnoses mirror	trigger (SECURITY DEFINER)
sync_lab_orders_from_tests() / sync_radiology_orders_from_tests()	triggers	Order creation from encounters.tests	triggers
tag_encounter_demographics() / process_encounter_indicators() / tag_encounter_from_room_visit()	triggers	MOH indicator tagging	triggers
get_user_display_name(p_user_id uuid)	→ text	Audit display names	authenticated

Section 6 — Webhook / Trigger Reference

Database triggers (all defined in supabase/migrations/; audited tables each carry INSERT/UPDATE/DELETE audit_* triggers via audit_trigger_fn()):

Trigger	Table	Event	Function
trg_set_sha_fund_type	encounters	BEFORE INSERT	set_sha_fund_type()
trg_auto_generate_sha_claim	encounters	AFTER UPDATE OF status	auto_generate_sha_claim()
trg_enforce_insurance_visit_limit	encounters	BEFORE INSERT OR UPDATE OF insurance_covered,	enforce_insurance_visit_limit
		insurance_pool_id,	
trg_shr_access_notification	encounters	AFTER INSERT	trg_shr_access_notification
trg_sync_encounter_diagnoses	encounters	AFTER INSERT OR UPDATE	sync_encounter_diagnoses
trg_sync_lab_orders_from_tests	encounters	AFTER INSERT OR UPDATE OF tests	sync_lab_orders_from_tests

Trigger	Table	Event	Function
trg_route_reg_to_lab / trg_route_registration_to_lab_ins	encounters	INSERT/UPDATE	route_registration_to_lab()
zzz_route_billing_ins / zzz_route_billing_upd	encounters	INSERT/UPDATE	route_registration_billing()
trg_route_registration_to_service_room	encounters	INSERT/UPDATE	route_registration_to_servi
trg_enforce_encounter_lock	encounter_diagnoses, clinical_notes, radiology_orders trg_enforce_encounter_lock_diagnoses	BEFORE INSERT/UPDATE/DELETE	enforce_encounter_lock()
trg_medication_dispense_fhir	prescriptions	AFTER UPDATE	trg_medication_dispense_fl
trg_route_prescription_to_pharmacy	prescriptions	INSERT/UPDATE	route_prescription_to_pharm
trg_dispense_prescription_stock	prescriptions	UPDATE	dispense_prescription_stoc
trg_shr_transmission_log	dha_outbound_queue	AFTER UPDATE / AFTER INSERT	trg_insert_transmission_log()
trg_shr_transmission_log_insert trg_smov_apply	stock_movements	AFTER INSERT OR DELETE	apply_stock_movement()
trg_deliv_stock	deliveries	AFTER INSERT	delivery_to_stock()
patient_registrations_insert_trg patient_registrations_update_trg	patient_registrations (view)	INSTEAD OF INSERT/UPDATE	patient_registrations_instead
audit_* (20 tables × 3)	clinical, financial, security, stock tables	AFTER	audit_trigger_fn()
set_*_updated_at family	app_settings, rooms, insurance_providers, lab_orders, lab_results, lab_test_catalog, profiles, mortuary_records, contracted_prices, test_templates, machines, machine_logs, deliveries, fund_utilizations, stock_items, insurance_transactions	INSERT/UPDATE/DELETE BEFORE UPDATE	family_updated_at()
trg_assign_mortuary_reference	mortuary_records	BEFORE INSERT	assign_mortuary_reference
sync_bed_status_on_admission	admissions	INSERT/UPDATE	sync_bed_status_on_admis
trg_tag_demographics / trg_process_moh_indicators	encounters	INSERT/UPDATE	tag_encounter_demograph process_encounter_indicate
trg_set_patient_file_number	patients	BEFORE INSERT	set_patient_file_number()

External webhooks: none inbound at present. ☐ PENDING (planned):
mpesa-callback webhook (Safaricom Daraja) and SHA/DHA asynchronous claim
status callbacks (per docs/integration-activation-manual.md).

End of DOC-8.